



CHALLENGE 11 / 20

THE COMPETITOR'S STOLEN FORMULA

LEAK-SOURCE ATTRIBUTION THROUGH TRANSFORMED DOCUMENTS

NATIONAL LEVEL HACKATHON • 2026

POWERED BY SYNTHOQUEST PVT LTD

Student Project Development Challenge • No organizer-provided evidence package.

1 | PROBLEM BACKGROUND

A specialty chemicals company holds a 63-page master formulation: process parameters, ratios, impurity control, catalyst treatment, tables, flow diagrams and five embedded spreadsheets. It exists in 31 versions and derivatives: internal edits, Hindi and Japanese translations, PDF exports and slide extracts. It went to 17 staff, four partners and two regulator portals. No watermark was embedded.

Eight months later a competitor filed a dossier for a near-identical product. The text is reworded, units are converted, tables are images, figures are redrawn and steps are merged or reordered. About 18% of numerical parameters have been deliberately perturbed.

The competitor claims independent development and cites public patents that overlap 40% of the content. The company must prove derivation to the standard of a court, and show which version and recipient lineage was the source.

2 | PROBLEM STATEMENT

Develop a **Derived-Document Lineage and Leak Attribution System** that establishes derivation across text, tables, figures, units, languages and deliberate perturbation, separates what could be obtained from public sources from what only the confidential document contained, and narrows the source to a version and recipient lineage using only intrinsic features.

The system must quantify the probability that the competitor's document arose independently, in a form an expert witness could defend.

Hard constraints

- Numbers are perturbed, units converted, text translated and tables rasterised.
- Overlap with public literature must be separated from confidential-only content.
- Fingerprint-like features such as errors and formatting quirks may be coincidental and must be treated as such.
- Independent-origin probability must be estimated and its method defended.
- The system must handle several media types and must not require the leaked document's source.

No evidence package or dataset is provided. Teams decide what to build, what data they need, how to simulate the environment and how to validate the result. No tools or architecture are prescribed.

3 | MANDATORY FEATURES

Teams must develop a working software solution. The system must provide the following features. How each is achieved is for the team to decide.

01	Multimodal Document Representation	Ability to represent text, tables, figures and images comparably.
02	Perturbation-Robust Matching	Ability to match content despite value changes and unit conversion.
03	Cross-Language Derivation Detection	Ability to detect derivation across translation.
04	Public-Source Separation	Ability to separate public-source content from confidential-only content.
05	Version Lineage Inference	Ability to place a leak within a tree of versions.
06	Recipient Narrowing	Ability to narrow the source to recipient lineages without pre-embedded marks.
07	Independent-Origin Estimation	Ability to estimate the probability of independent creation.
08	Expert-Defensible Reporting	Ability to present findings suitable for expert testimony.

The system must clearly distinguish between:

IDENTICAL	DERIVED	PARTIALLY REUSED	COINCIDENTAL	UNDETERMINED
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A derivation claim that cannot be supported must not be presented as proof of copying.

Build, don't search. This is a development challenge. Teams must create the system and the controlled environment that proves it works. Submitting an analysis of a given dataset is not a solution.

4 | EXPECTED OUTPUT

Live demonstration. The team must show that its system can handle a complex controlled scenario created by the team. The demonstration must show:

01	Teams' own simulated confidential document family with versions and recipients
02	A leak transformed by rewording, units, translation and image conversion
03	Detection of derivation despite perturbed values
04	Separation of public-source content
05	Version and recipient narrowing
06	Independent-origin probability
07	Findings an expert witness could present

The team must be able to explain why the system reached each major conclusion.

Validation standard (every team)

- Build a scenario generator with known ground truth and evaluate on at least ten distinct scenarios, including ones with deliberately misleading evidence.
- At the demonstration, judges change scenario parameters on the spot. The system must adapt without being rebuilt.
- Include an adversary variant that actively tries to defeat the system, and report how the system held up.
- Stated confidence must match measured accuracy. Overconfident or unexplained conclusions lose marks.
- No real systems, real personal data or functional malware may be used.

Submission. Each team must submit:

01	Working Prototype	A functional implementation of the proposed solution.
02	Source Code	Complete source code required to reproduce the project.
03	Technical Architecture	Architecture and component description.
04	Validation Environment	A controlled environment created by the team to demonstrate the solution.
05	Technical Documentation	Implementation, design and testing documentation.
06	Forensic Investigation Report	A professional report generated from the demonstrated investigation.
07	Demonstration	Live technical demonstration to the judging panel.
08	Limitations	Known limitations and situations where the system cannot establish a reliable conclusion.